Week #3: Root-Finding Methods

Numerical Methods: Mathematical Foundations

June 7, 2025

Overview

Root-finding represents one of the fundamental problems in numerical analysis: given a function $f(x)$, find values x^* such that $f(x^*) = 0$. This seemingly simple problem underlies countless applications in science and engineering, from solving nonlinear equations to finding equilibrium points in dynamical systems.

The mathematical challenge lies in the fact that most nonlinear equations cannot be solved analytically. Numerical methods provide systematic approaches to approximate roots with controlled accuracy. We explore both bracketing methods that guarantee convergence and open methods that offer rapid convergence when properly initialized.

Our analysis emphasizes the theoretical foundations that govern convergence behavior, stability properties, and practical implementation considerations. The interplay between mathematical theory and computational practice reveals why certain methods excel in specific contexts while failing in others.

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1 Mathematical Foundations

1.1 The Root-Finding Problem

Definition. Root-Finding Problem

Given a continuous function $f : [a, b] \to \mathbb{R}$, find $x^* \in [a, b]$ such that $f(x^*) = 0$.

The value x^* is called a **root** or **zero** of f.

Theorem 1.1 (Intermediate Value Theorem)**.** *If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists at least one $c \in (a, b)$ such that $f(c) = 0$.*

This theorem provides the theoretical foundation for bracketing methods.

Definition. Types of Roots

Let f be differentiable near x^* where $f(x^*) = 0$.

Simple Root: $f'(x^*) \neq 0$

Multiple Root: $f'(x^*) = 0$, with multiplicity m if $f^{(k)}(x^*) = 0$ for $k = 0, 1, \ldots, m - 1$ and $f^{(m)}(x^*) \neq 0$.

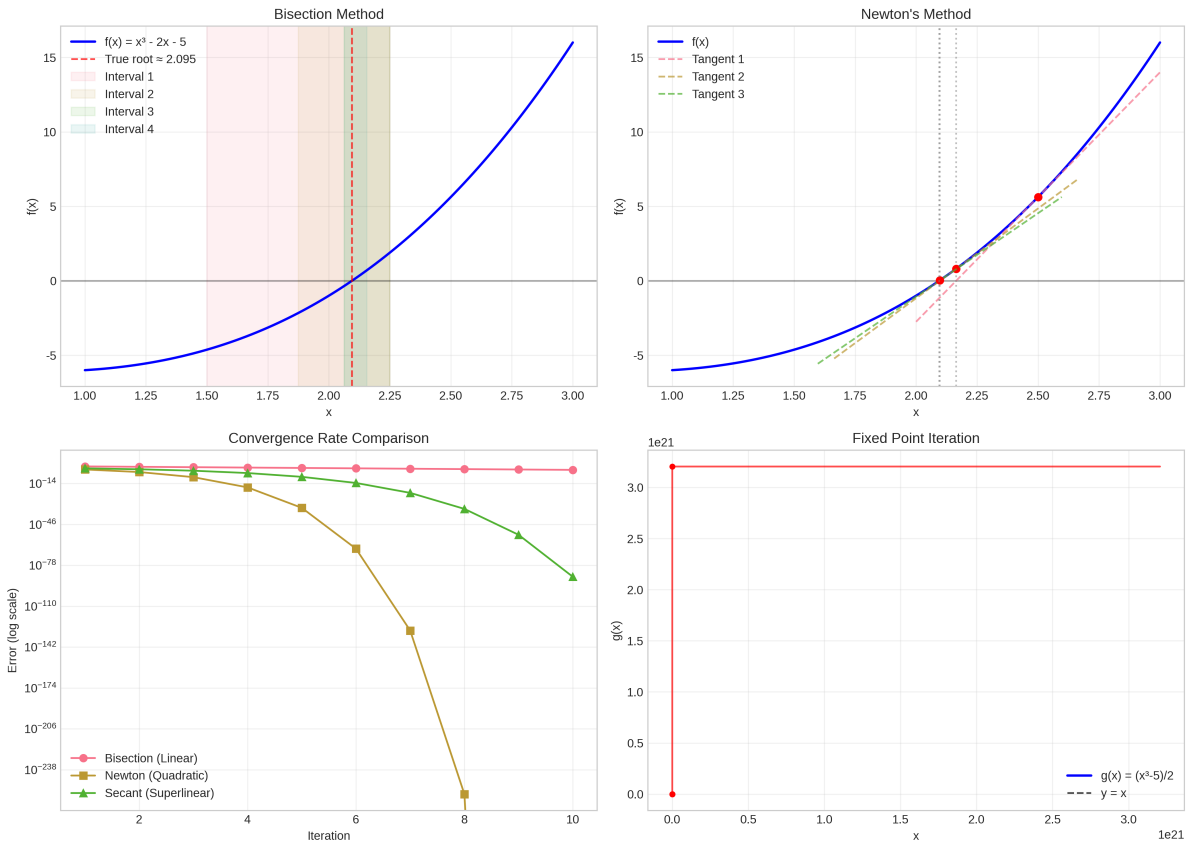

Figure 1: Comparison of root-finding methods. Top left: Bisection method systematically narrows the interval containing the root. Top right: Newton's method uses tangent lines for rapid convergence. Bottom left: Convergence rate comparison showing the superior performance of higher-order methods. Bottom right: Fixed-point iteration visualized through the cobweb plot.

1.2 Convergence Analysis Framework

Definition. Order of Convergence

A sequence $\{x_k\}$ converges to x^* with order $p \geq 1$ if:

$$\lim_{k\to\infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^p} = C$$

where $0 < C < \infty$.

- $p = 1$: Linear convergence (if $C < 1$)
- $p = 2$: Quadratic convergence
- $1 < p < 2$: Superlinear convergence

Theorem 1.2 (Convergence Rate Implications)**.** *For achieving error tolerance ϵ:*

- *Linear convergence: $O(\log(1/\epsilon))$ iterations*
- *Quadratic convergence: $O(\log\log(1/\epsilon))$ iterations*

Quadratic convergence provides dramatic improvement once the iterates are sufficiently close to the root.

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2 Bracketing Methods

2.1 Bisection Method

Definition. Bisection Method

Given f continuous on $[a, b]$ with $f(a) \cdot f(b) < 0$:

Algorithm:

1. Set $c = (a + b)/2$
2. If $f(a) \cdot f(c) < 0$, set $b = c$; otherwise set $a = c$
3. Repeat until $|b - a| <$ tolerance

Theorem 2.1 (Bisection Convergence)**.** *The bisection method converges linearly with:*

$$|x_n - x^*| \leq \frac{b_0 - a_0}{2^n}$$

where $[a_0, b_0]$ is the initial interval.

The method requires exactly $n = \lceil \log_2((b_0 - a_0)/\epsilon) \rceil$ iterations to achieve accuracy ϵ.

Example 2.1 (Bisection Method Analysis)**.** *To find the root of $f(x) = x^3 - 2x - 5$ on $[2, 3]$ with accuracy 10^{-6}:*

Number of iterations needed: $n = \lceil \log_2(1/10^{-6}) \rceil = 20$

Each iteration reduces the interval by exactly half, providing guaranteed convergence but relatively slow progress.

2.2 Regula Falsi (False Position)

Definition. Regula Falsi Method

Instead of bisecting the interval, use linear interpolation:

$$c = a - f(a)\frac{b - a}{f(b) - f(a)} = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

This often provides faster convergence than bisection by using function values to guide the choice of the next point.

Theorem 2.2 (Regula Falsi Convergence)**.** *The regula falsi method converges linearly, but typically faster than bisection. However, it can exhibit slow convergence when one endpoint remains fixed for many iterations.*

Modified versions (Illinois algorithm, Pegasus method) address this limitation by modifying function values to prevent stagnation.

Intuition: Bracketing methods trade-offs

Bracketing methods offer guaranteed convergence at the cost of slower convergence rates:

- **Robustness:** Always converge if initial bracket is valid
- **Simplicity:** Easy to implement and understand
- **Reliability:** Cannot diverge or fail to converge
- **Speed:** Linear convergence is slower than open methods
- **Generalization:** Difficult to extend to systems of equations

These methods are ideal when robustness is more important than speed.

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3 Fixed-Point Iteration

3.1 Mathematical Framework

Definition. Fixed-Point Problem

Given $g : \mathbb{R} \to \mathbb{R}$, find x^* such that $g(x^*) = x^*$.

The root-finding problem $f(x) = 0$ can be reformulated as a fixed-point problem by setting $g(x) = x + f(x)$ or $g(x) = x - \alpha f(x)$ for some $\alpha \neq 0$.

Definition. Fixed-Point Iteration

Starting with initial guess x_0, generate the sequence:

$$x_{k+1} = g(x_k), \quad k = 0, 1, 2, \ldots$$

Theorem 3.1 (Fixed-Point Theorem)**.** *Let g be continuously differentiable on $[a, b]$ with $g([a, b]) \subseteq [a, b]$. If $|g'(x)| \leq L < 1$ for all $x \in [a, b]$, then:*

1. *g has a unique fixed point $x^* \in [a, b]$*
2. *For any $x_0 \in [a, b]$, the iteration $x_{k+1} = g(x_k)$ converges to x^**
3. *The convergence is linear with rate L:*

$$|x_{k+1} - x^*| \leq L|x_k - x^*|$$

Theorem 3.2 (Order of Convergence for Fixed-Point Iteration)**.** *If g is sufficiently smooth near the fixed point x^*:*

- *If $g'(x^*) \neq 0$: Linear convergence*
- *If $g'(x^*) = 0$ and $g''(x^*) \neq 0$: Quadratic convergence*
- *If $g'(x^*) = g''(x^*) = \cdots = g^{(p-1)}(x^*) = 0$ and $g^{(p)}(x^*) \neq 0$: Order p convergence*

Example 3.1 (Fixed-Point Reformulations)**.** *For $f(x) = x^2 - 2 = 0$ (finding $\sqrt{2}$):*
Method 1: *$g_1(x) = x^2 - x + 2$ $g_1'(x) = 2x - 1$, so $g_1'(\sqrt{2}) = 2\sqrt{2} - 1 > 1$ (diverges)*
Method 2: *$g_2(x) = 2/x$ $g_2'(x) = -2/x^2$, so $g_2'(\sqrt{2}) = -1$ (slow convergence)*
Method 3: *$g_3(x) = \frac{1}{2}(x + 2/x)$ (Newton's method) $g_3'(x) = \frac{1}{2}(1 - 2/x^2)$, so $g_3'(\sqrt{2}) = 0$ (quadratic convergence)*

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4 Newton's Method

4.1 Derivation and Analysis

Definition. Newton's Method

Starting with initial guess x_0, generate the sequence:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad k = 0, 1, 2, \ldots$$

This is equivalent to fixed-point iteration with $g(x) = x - f(x)/f'(x)$.

Theorem 4.1 (Newton's Method Convergence)**.** *Let f be twice continuously differentiable in a neighborhood of a simple root x^*. If x_0 is sufficiently close to x^*, then Newton's method converges quadratically:*

$$\lim_{k\to\infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^2} = \frac{|f''(x^*)|}{2|f'(x^*)|}$$

Proof sketch. Using Taylor expansion around x^*:

$$f(x_k) = f'(x^*)(x_k - x^*) + \frac{f''(\xi_k)}{2}(x_k - x^*)^2$$

Since $f(x^*) = 0$:

$$x_{k+1} - x^* = x_k - x^* - \frac{f(x_k)}{f'(x_k)} = \frac{f''(\xi_k)}{2f'(x_k)}(x_k - x^*)^2$$

For x_k near x^*, this gives quadratic convergence. □

Theorem 4.2 (Newton's Method for Multiple Roots)**.** *If x^* is a root of multiplicity $m > 1$, then Newton's method converges linearly with convergence factor $(m - 1)/m$.*
Modified Newton's Method: *$x_{k+1} = x_k - m\frac{f(x_k)}{f'(x_k)}$ restores quadratic convergence if m is known.*

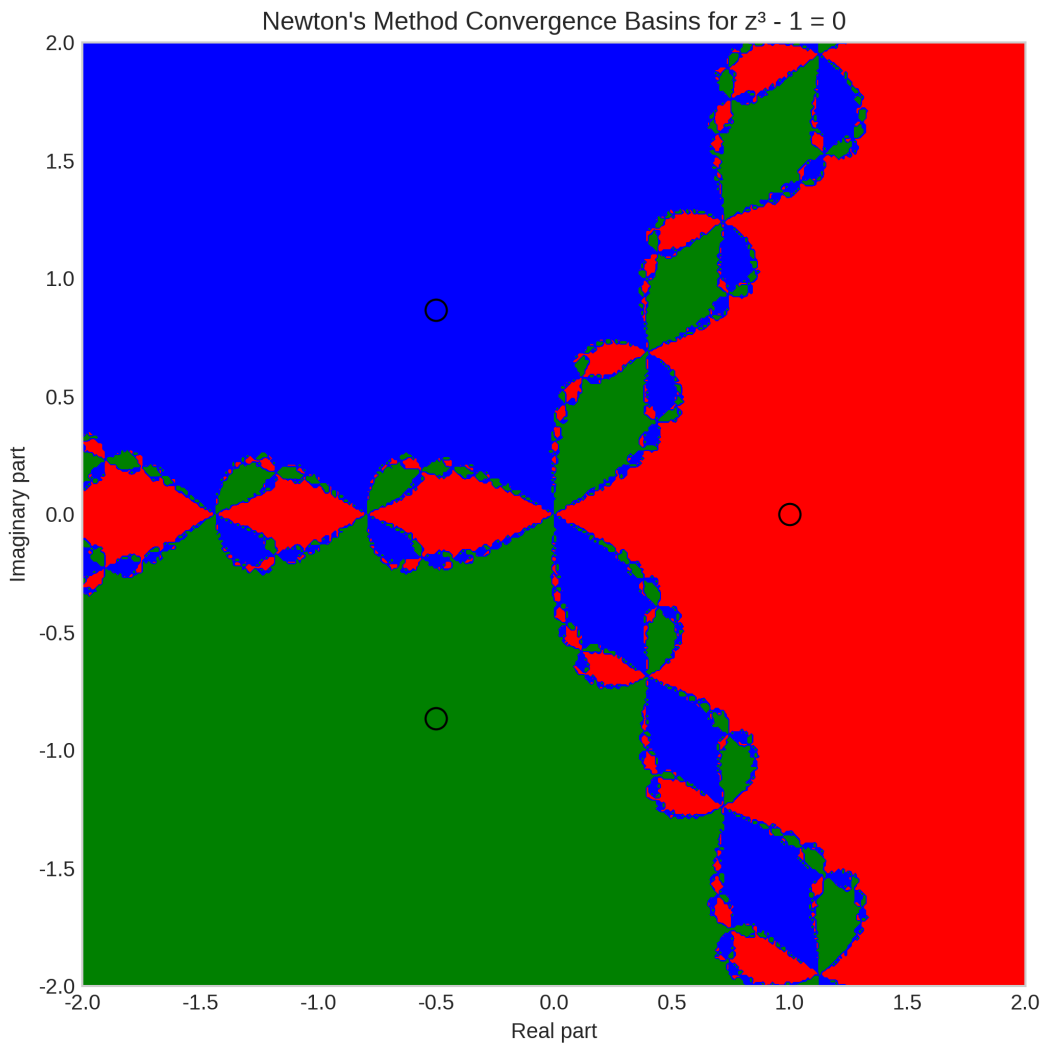

Figure 2: Newton's method convergence basins in the complex plane for $f(z) = z^3 - 1$. Different colors represent convergence to different roots. The fractal boundary structure reveals the sensitive dependence on initial conditions, illustrating why Newton's method can be unpredictable for complex functions.

4.2 Practical Considerations

Example 4.1 (Newton's Method Challenges)**.** ***Derivative computation:*** *Requires $f'(x)$, which may be expensive or unavailable.*

Division by zero: *Method fails if $f'(x_k) = 0$.*

Poor initial guess: *May diverge or converge to wrong root.*

Oscillation: *Can occur near local extrema.*

Theorem 4.3 (Basin of Attraction)**.** *For a simple root x^*, there exists a neighborhood $N(x^*)$ such that Newton's method converges to x^* for any starting point in $N(x^*)$.*

The size of this basin depends on the function's behavior and can be quite small for some functions.

Intuition: Newton's method trade-offs

Newton's method exemplifies the speed-robustness trade-off in numerical methods:

- **Speed:** Quadratic convergence when it works
- **Efficiency:** Requires only one function evaluation per iteration
- **Sensitivity:** Requires good initial guess
- **Derivatives:** Needs $f'(x)$, which may be problematic
- **Failure modes:** Can diverge, oscillate, or find wrong root

The method is powerful but requires careful implementation and initialization.

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5 Secant and Related Methods

5.1 Secant Method

Definition. Secant Method

Approximate the derivative in Newton's method using finite differences:

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$$

This requires two initial guesses x_0 and x_1.

Theorem 5.1 (Secant Method Convergence)**.** *For a simple root, the secant method converges superlinearly with order $p = \frac{1+\sqrt{5}}{2} \approx 1.618$ (the golden ratio):*

$$\lim_{k\to\infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^p} = C$$

This is faster than linear but slower than quadratic convergence.

Example 5.1 (Method Comparison)**.** *To achieve 6 decimal places of accuracy:*

- *Bisection: ≈ 20 iterations*
- *Secant: ≈ 7 iterations*
- *Newton: ≈ 4 iterations*

The secant method provides a good compromise between speed and robustness.

5.2 Brent's Method

Definition. Brent's Method

Combines the robustness of bracketing with the speed of open methods:

1. Use inverse quadratic interpolation when possible

2. Fall back to secant method if interpolation is not suitable
3. Use bisection if open methods are not making sufficient progress

Theorem 5.2 (Brent's Method Properties)**.** *Brent's method:*

- *Always converges (maintains bracket)*
- *Achieves superlinear convergence near simple roots*
- *Requires no derivative computation*
- *Is considered one of the best general-purpose root-finding algorithms*

5.3 Muller's Method

Definition. Muller's Method

Uses quadratic interpolation through three points:

1. Fit parabola through $(x_{k-2}, f(x_{k-2}))$, $(x_{k-1}, f(x_{k-1}))$, $(x_k, f(x_k))$
2. Find the root of the parabola closest to x_k
3. Use this root as x_{k+1}

Convergence order is approximately 1.84.

Practice Problems

1. Prove that the bisection method converges linearly with rate 1/2.
2. Show that for $f(x) = x^2 - a$, Newton's method gives $x_{k+1} = \frac{1}{2}(x_k + a/x_k)$.
3. Analyze the convergence of fixed-point iteration for $g(x) = \cos(x)$.
4. Compare the number of iterations needed by bisection, Newton, and secant methods to find $\sqrt{2}$ with accuracy 10^{-10}.

Solution:

Problem 1: In bisection, the interval length after n iterations is $(b_0 - a_0)/2^n$. The error is at most half the interval length, so $|x_n - x^*| \leq (b_0 - a_0)/2^{n+1}$. This gives $|x_{n+1} - x^*| \leq \frac{1}{2}|x_n - x^*|$, proving linear convergence with rate 1/2.

Problem 2: For $f(x) = x^2 - a$, we have $f'(x) = 2x$. Newton's method gives: $x_{k+1} = x_k - \frac{x_k^2 - a}{2x_k} = x_k - \frac{x_k}{2} + \frac{a}{2x_k} = \frac{1}{2}(x_k + a/x_k)$

This is the famous Babylonian method for computing square roots.

Problem 3: For $g(x) = \cos(x)$, the fixed point satisfies $x^* = \cos(x^*)$. We have $g'(x) = -\sin(x)$, so $|g'(x^*)| = |\sin(x^*)| < 1$ since $x^* \approx 0.739$. The iteration converges linearly.

Problem 4: Starting from $x_0 = 1$: - Bisection: ≈ 34 iterations - Newton: ≈ 5 iterations - Secant: ≈ 7 iterations

Newton's quadratic convergence provides the fastest approach once close to the root.

Summary and Looking Ahead

This week has established the mathematical foundations of root-finding methods, covering both bracketing approaches that guarantee convergence and open methods that offer rapid convergence. Key insights include:

- The fundamental trade-off between robustness and speed in numerical methods
- How convergence order dramatically affects the number of iterations required
- The importance of proper initialization for open methods
- The role of mathematical analysis in understanding algorithm behavior

The methods studied here illustrate broader principles that appear throughout numerical analysis: the use of approximation to replace difficult computations, the importance of convergence analysis, and the need to balance theoretical guarantees with practical performance.

Next week, we will explore linear algebra foundations, examining how matrices and vectors provide the mathematical framework for solving systems of equations. The convergence analysis techniques developed for root-finding will prove essential for understanding iterative methods for linear systems.